



MEHARRY
SCHOOL OF APPLIED
COMPUTATIONAL SCIENCES

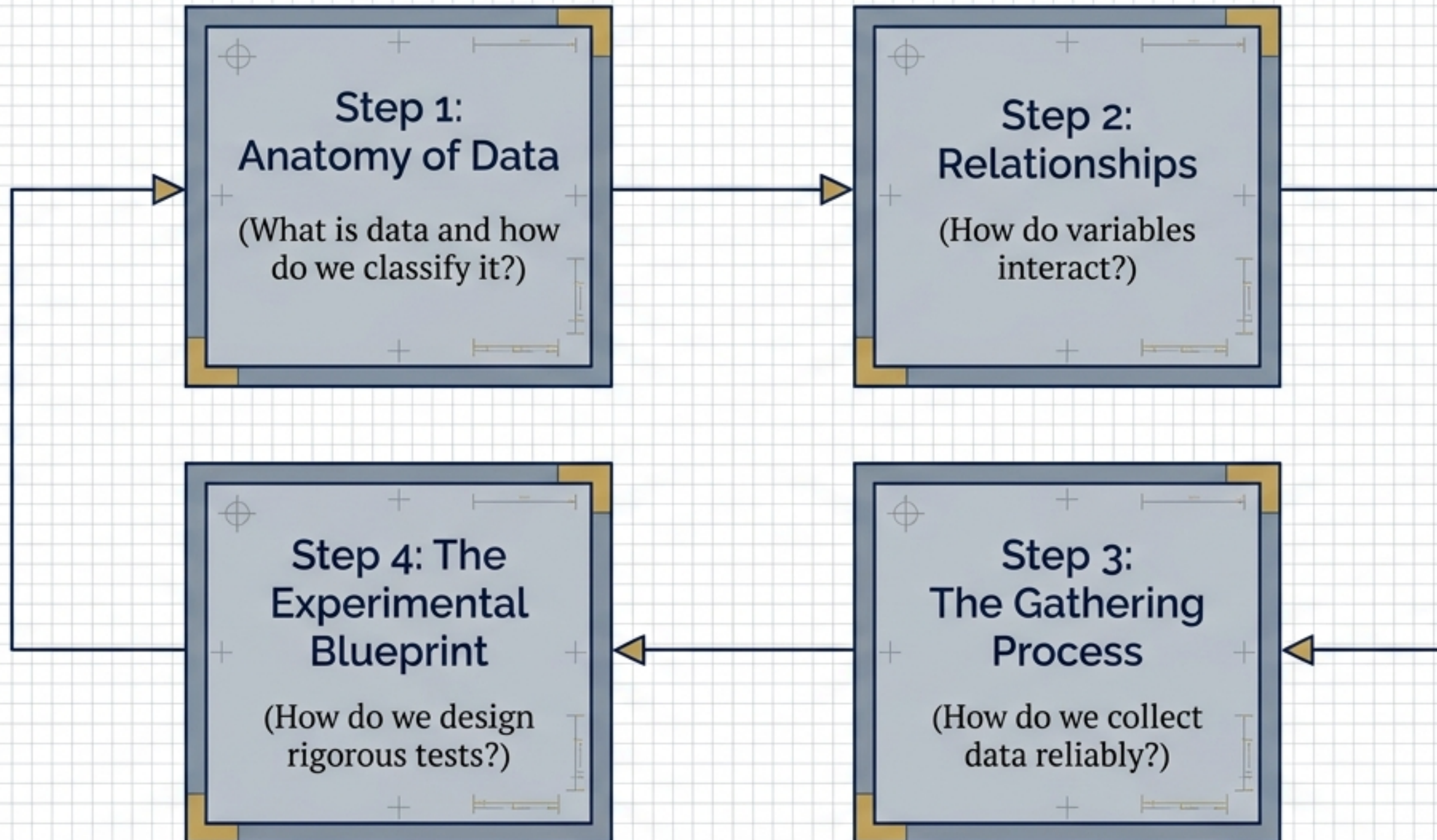
Introduction to Biostatistics

Course MSBD520

Fall 2026

Instructor: Nazirah Mohd Khairi, PhD

The Lifecycle of Inquiry



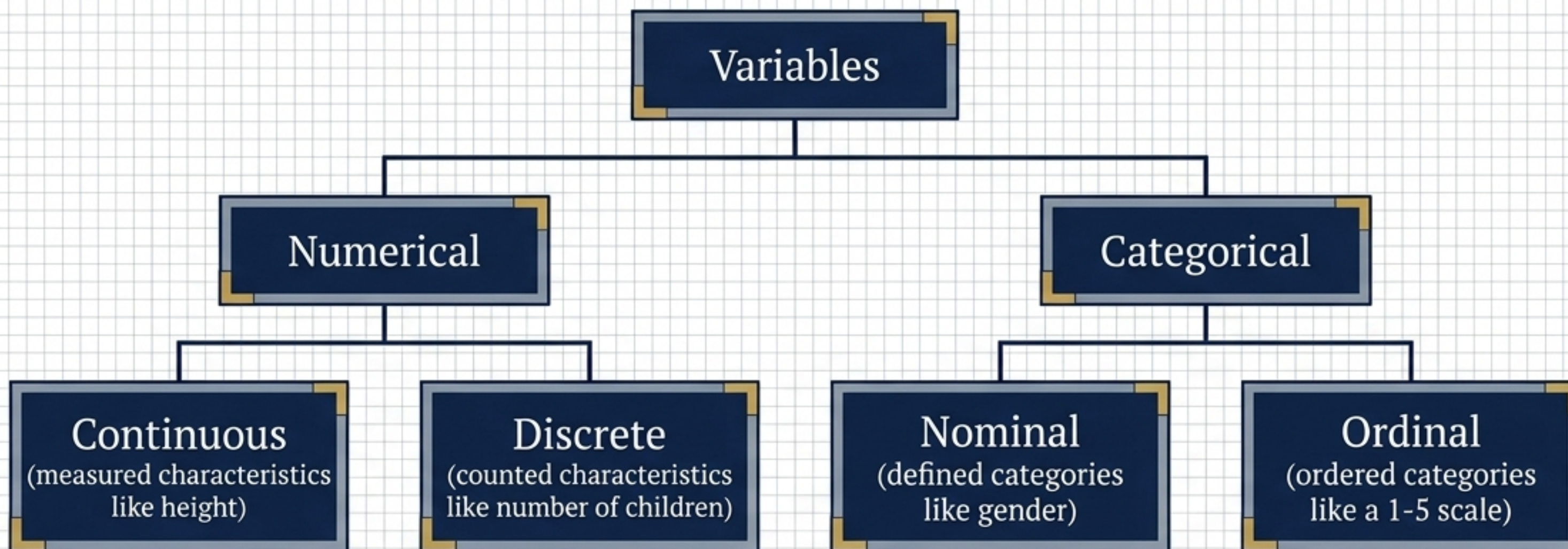
The Anatomy of a Data Matrix

gender	sleep	bedtime	countries	dread
male	7	11:00 PM	3	3
female	6	12:30 AM	1	2
female	8	10:00 PM	2	4
female	5	2:00 AM	4	2
...	...	...	...	...
male	7	11:30 PM	2	3

Variables: Recorded characteristics (e.g., hours of sleep, dread scale).

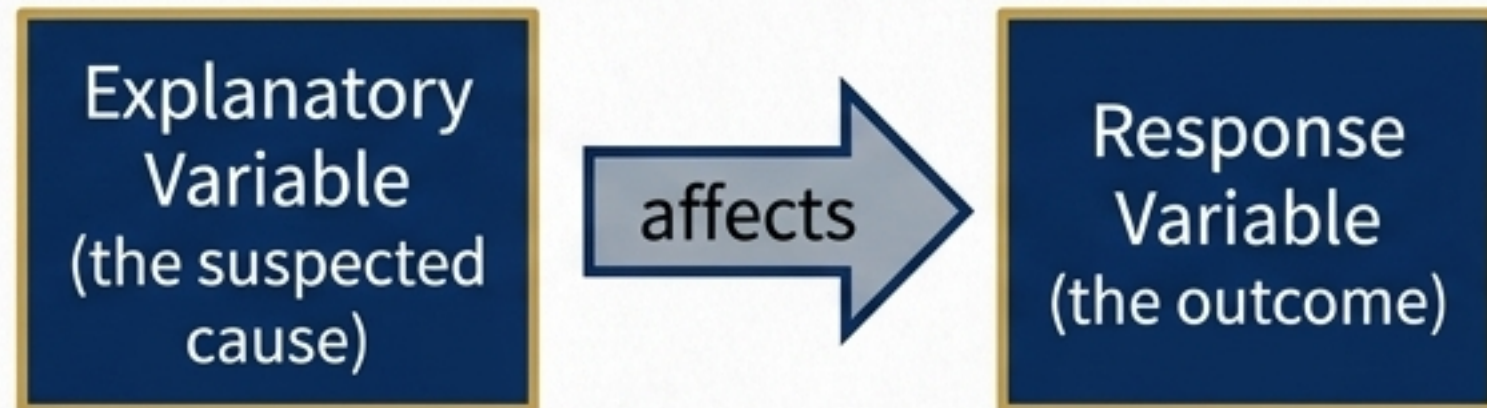
Observations: Individual data points (e.g., Student 1, Student 2).

The Taxonomy of Variables

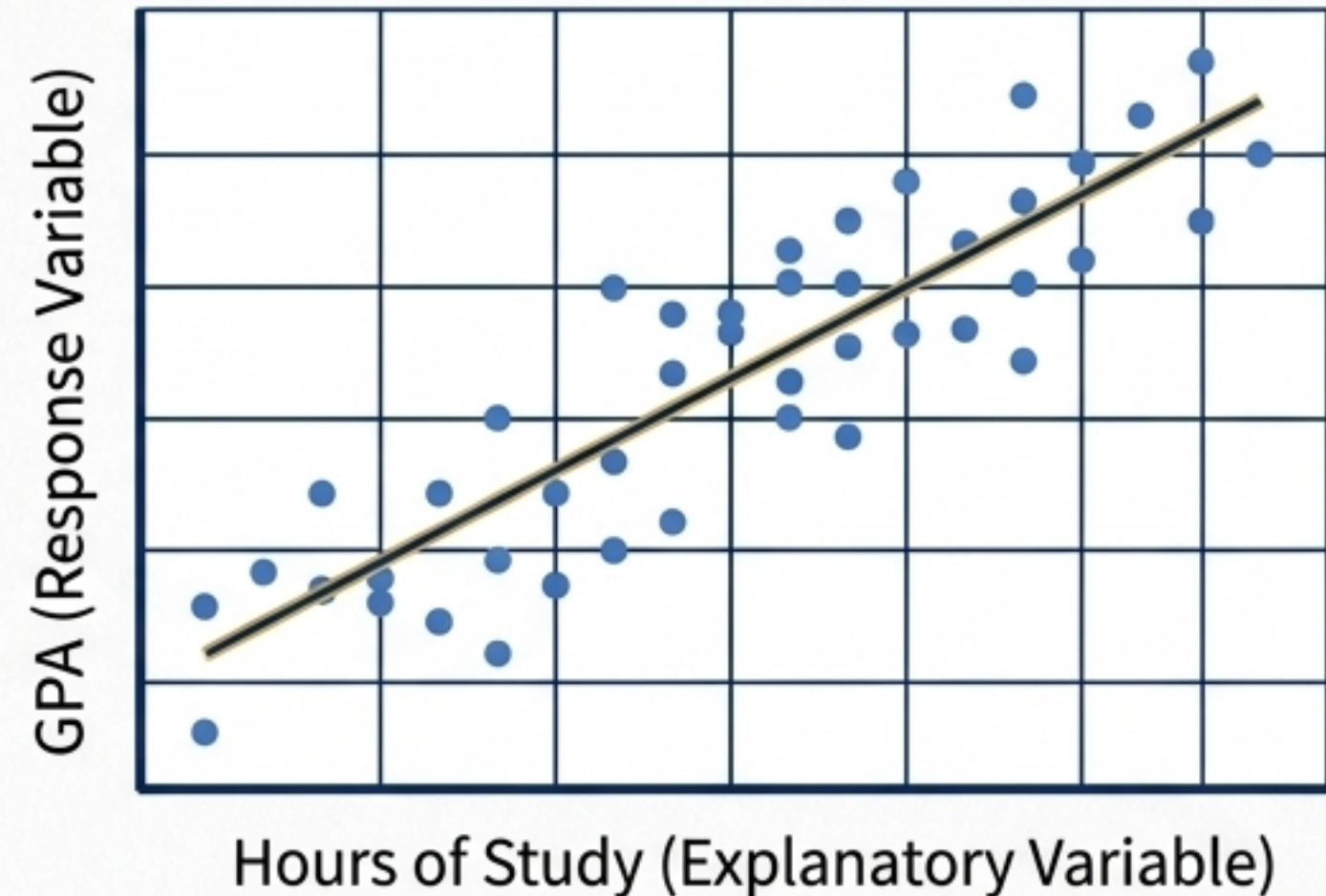


Is a telephone area code numerical?
No, it's categorical (nominal) because the numbers do not hold mathematical weight or inherent order.

Defining Relationships in Data



Example: In the frog maternal investment study, **Altitude** is the **Explanatory Variable**, and **Egg/Clutch Size** is the **Response Variable**.



Note: We use these labels to track suspicions; they do not guarantee actual causation.

Methods of Inquiry: Observational vs. Experimental

Observational Studies



Researchers collect data without interfering with how it arises. Establishes associations between variables, but cannot prove causation.

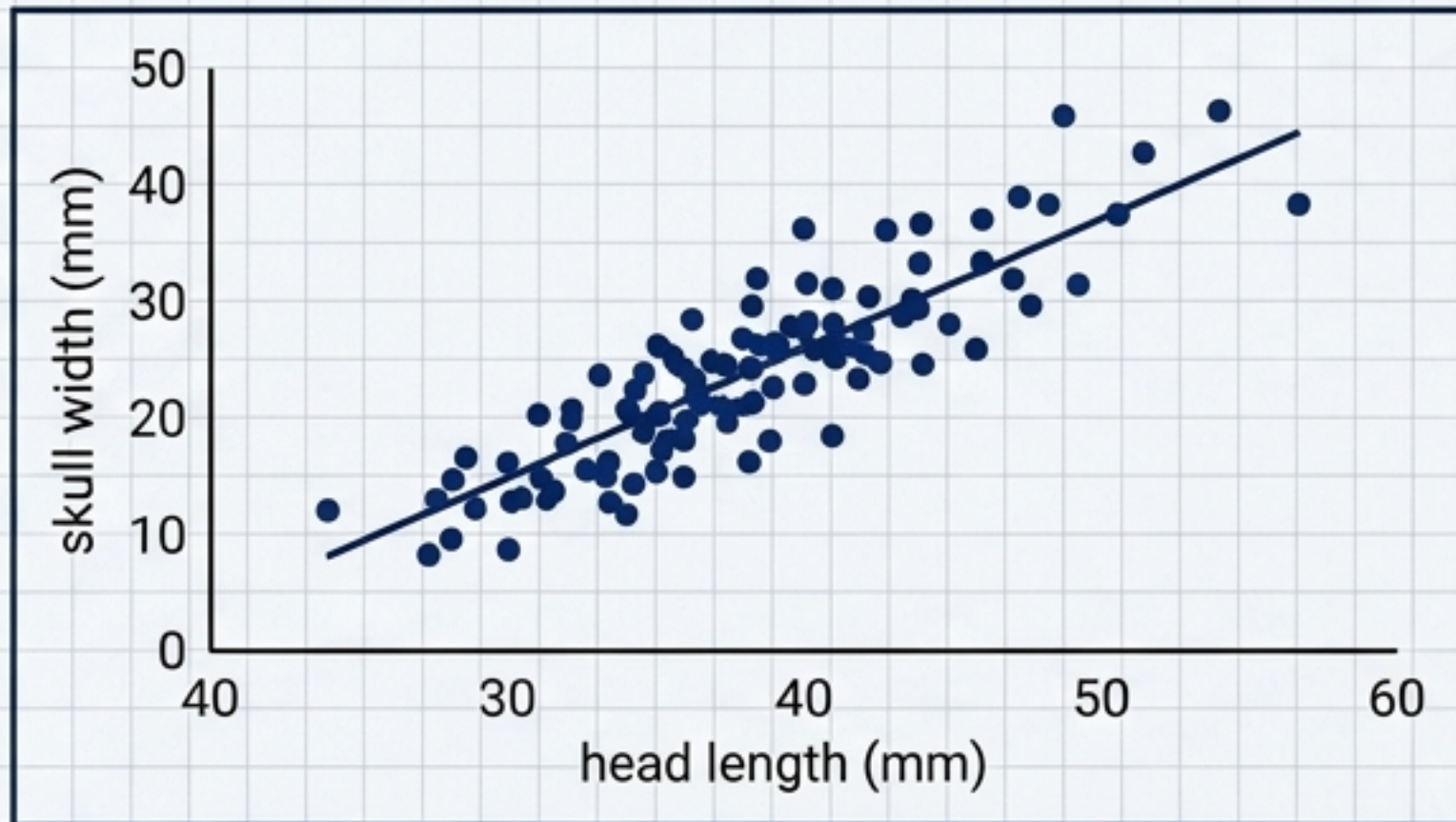
Experiments



Researchers randomly assign subjects to various treatments. Used to establish a direct causal connection between explanatory and response variables.

The Golden Rule

Association $\neq$ Causation.

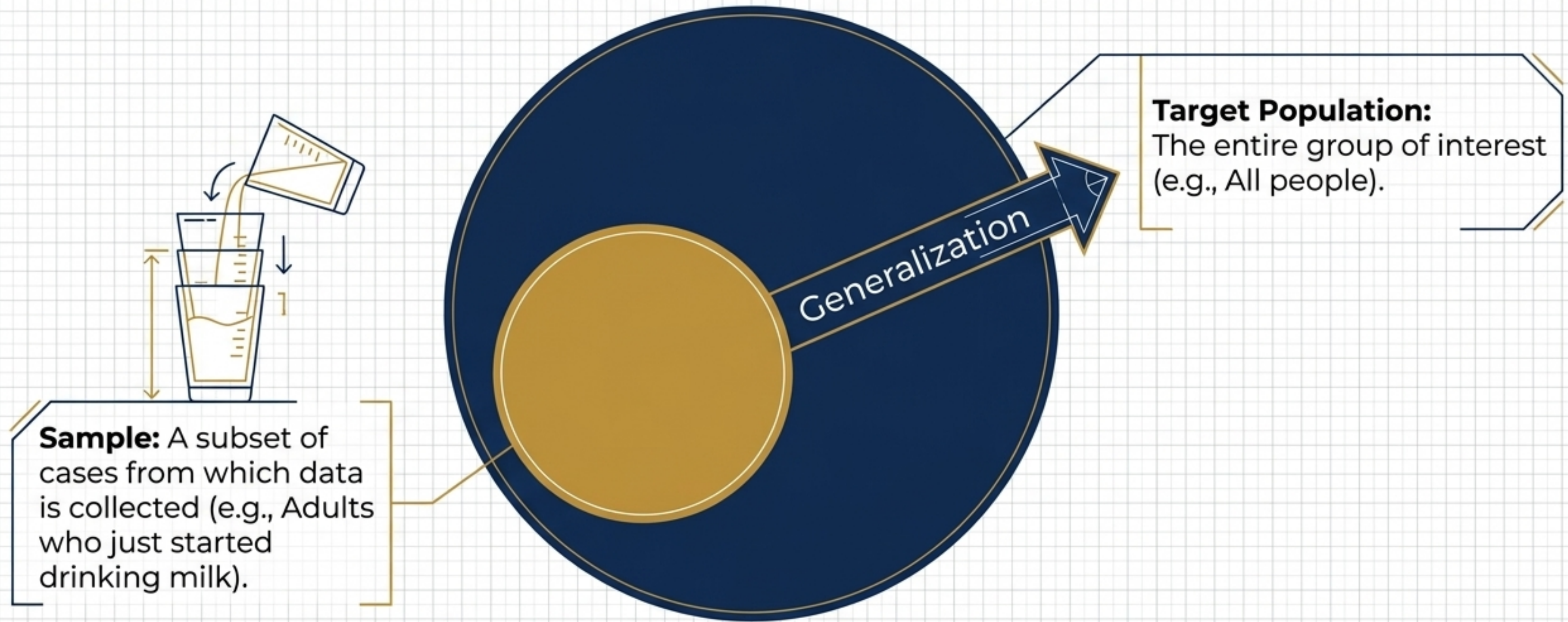


Data check: Possum head length and skull width are positively associated (dependent variables).

The Tralfamadore Thought Experiment

King Slartibartfast observed that people with longer names were taller. This is an observational study showing an association, but lengthening a name does not cause someone to grow taller.

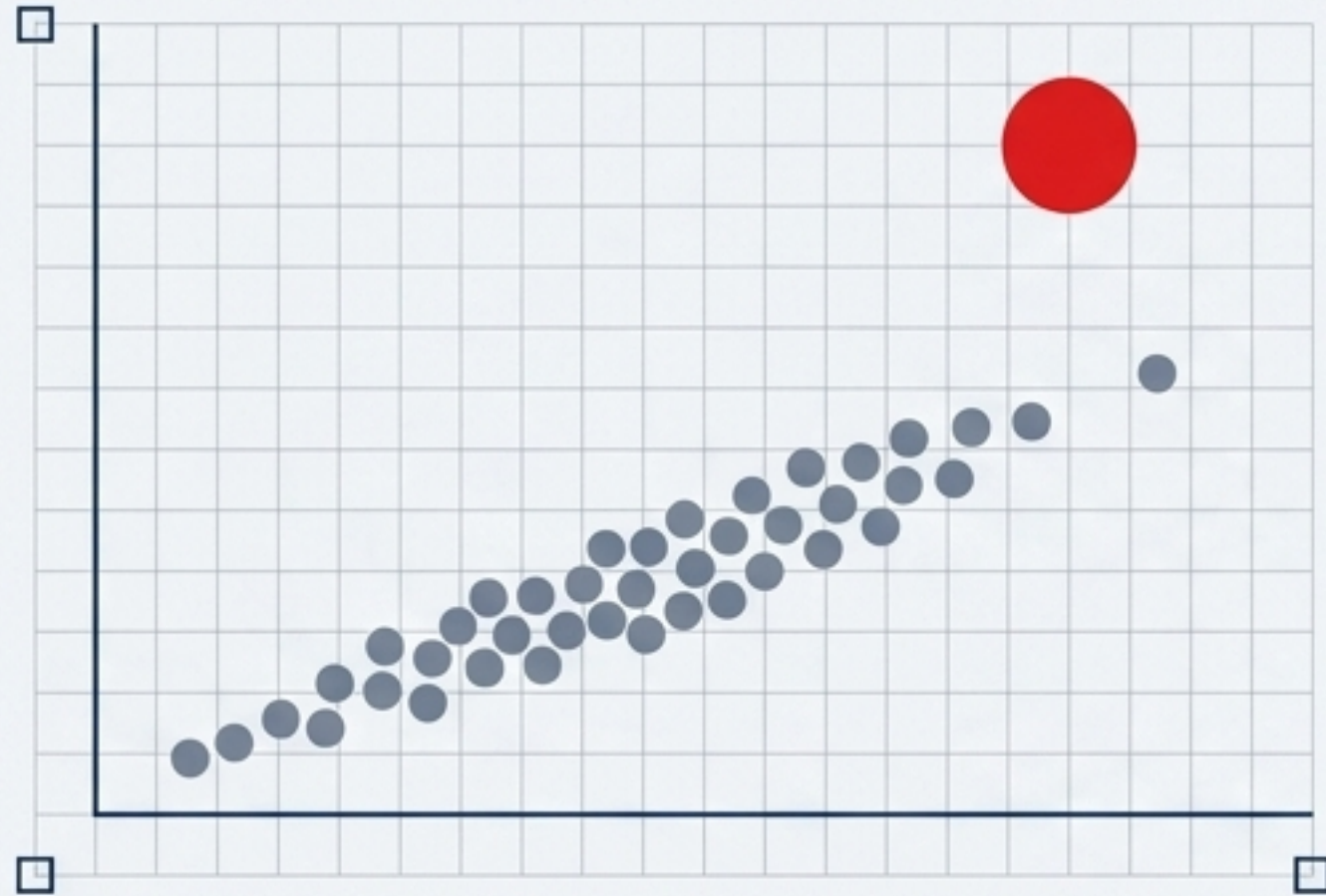
The Target: Populations vs. Samples



The Goal: A randomly selected sample allows us to generalize our findings back to the broader population.

The Pitfalls of Poor Data Gathering

Anecdotal Evidence



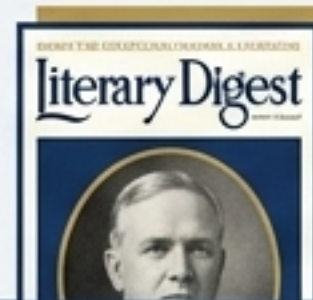
My uncle smokes 3 packs a day and is in perfect health.

Fails because a single haphazard data point ignores the reality of human variability.

Sampling Bias

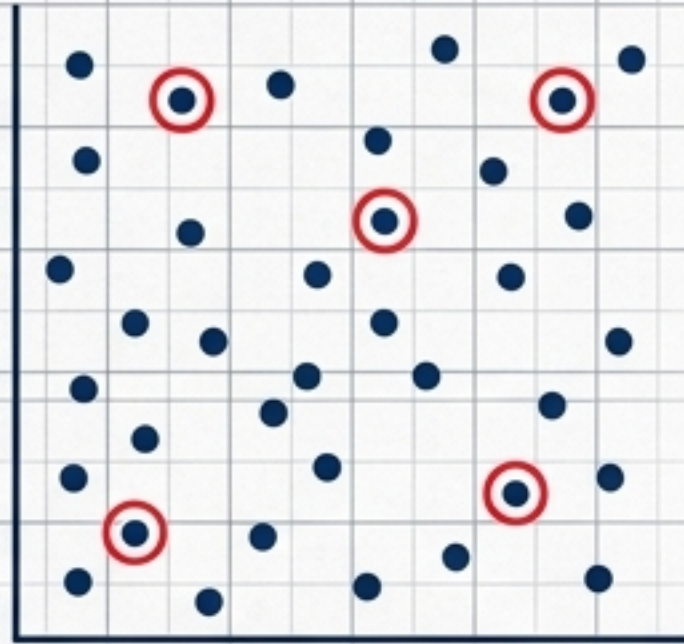


The Unstirred Soup: If a soup isn't stirred, a big spoon won't help you taste the whole recipe.



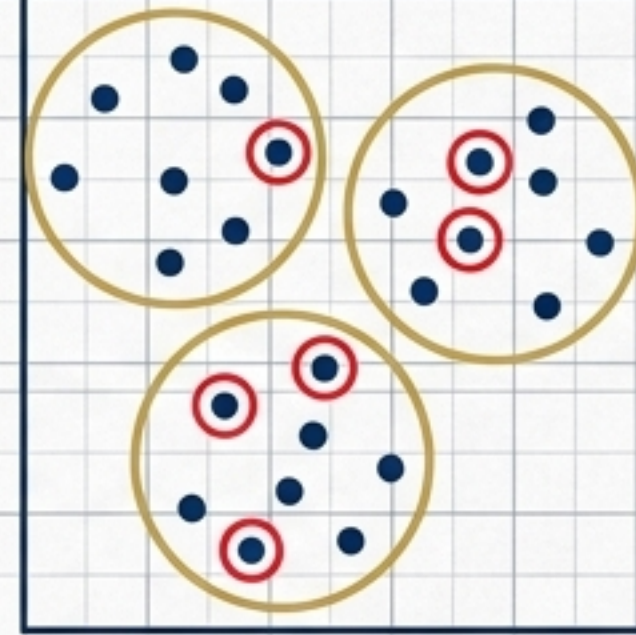
Example: The 1936 *Literary Digest* poll surveyed 2.4 million people but falsely predicted Landon over FDR because they only polled wealthy people with phones and cars during the Great Depression.

Four Pillars of Probability Sampling



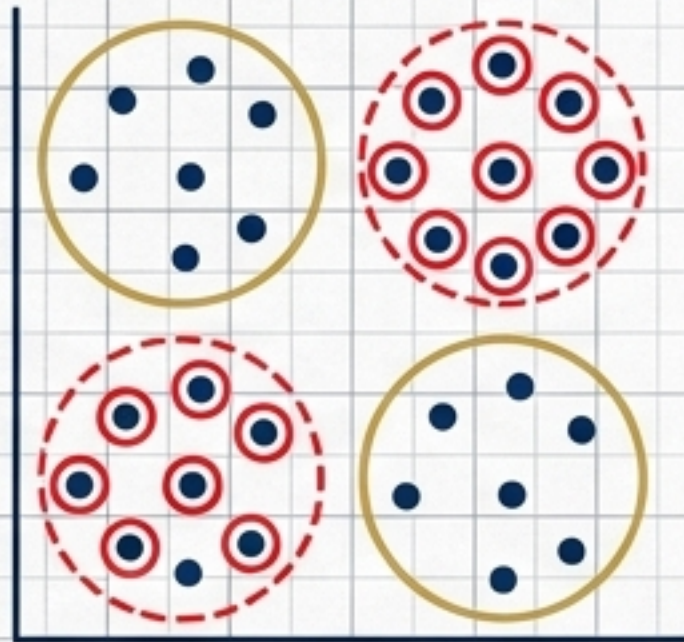
1. Simple Random

Cases selected with no implied connection.



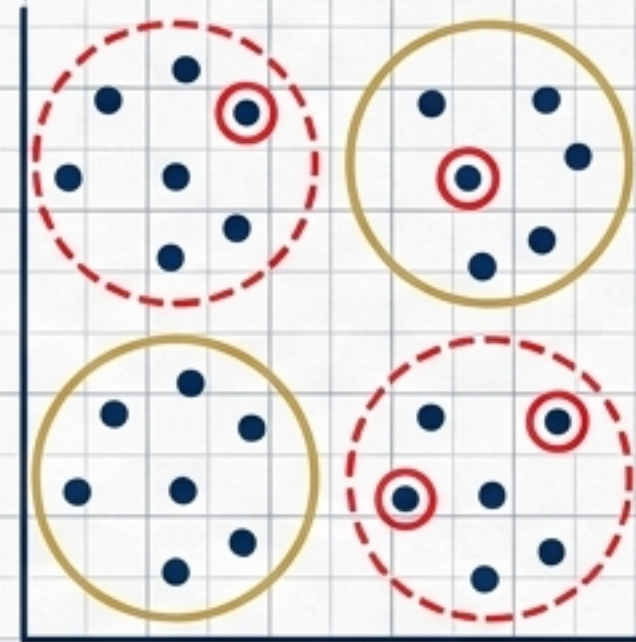
2. Stratified

Divide population into similar groups (strata), take a simple random sample from each.



3. Cluster

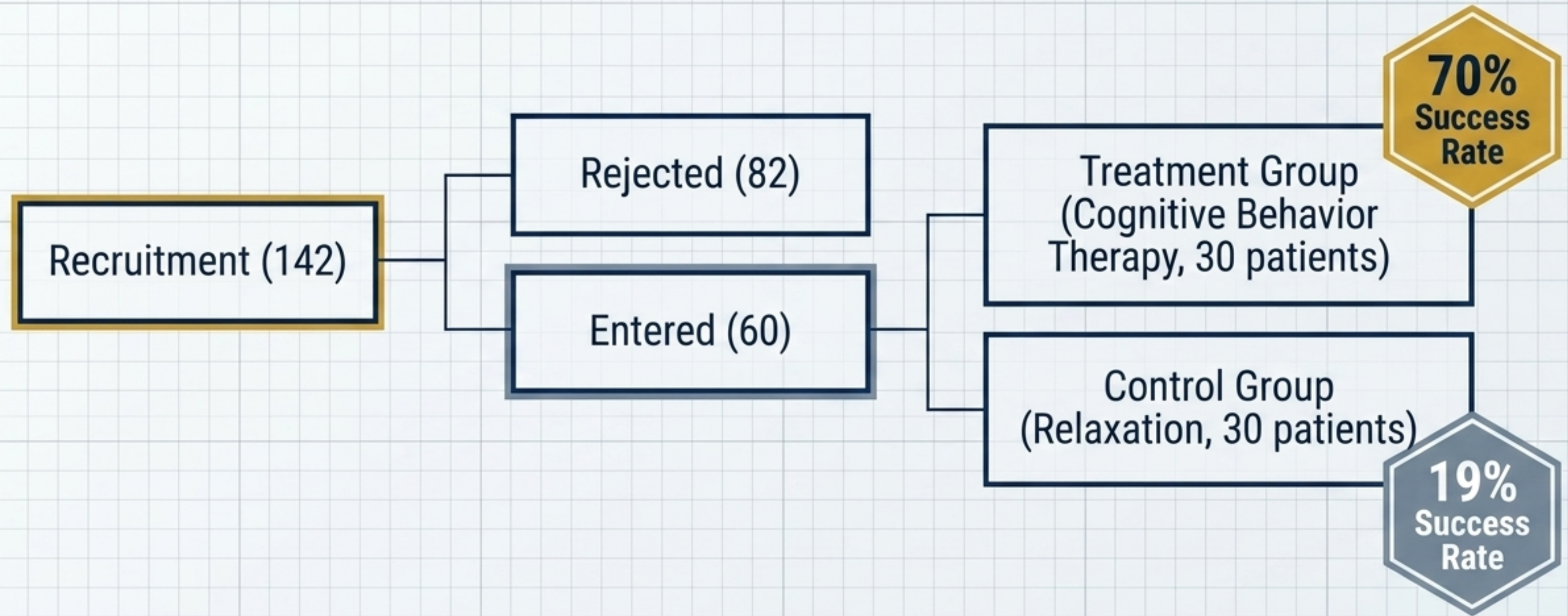
Divide into many groups, randomly select a few whole groups and sample everyone inside them.



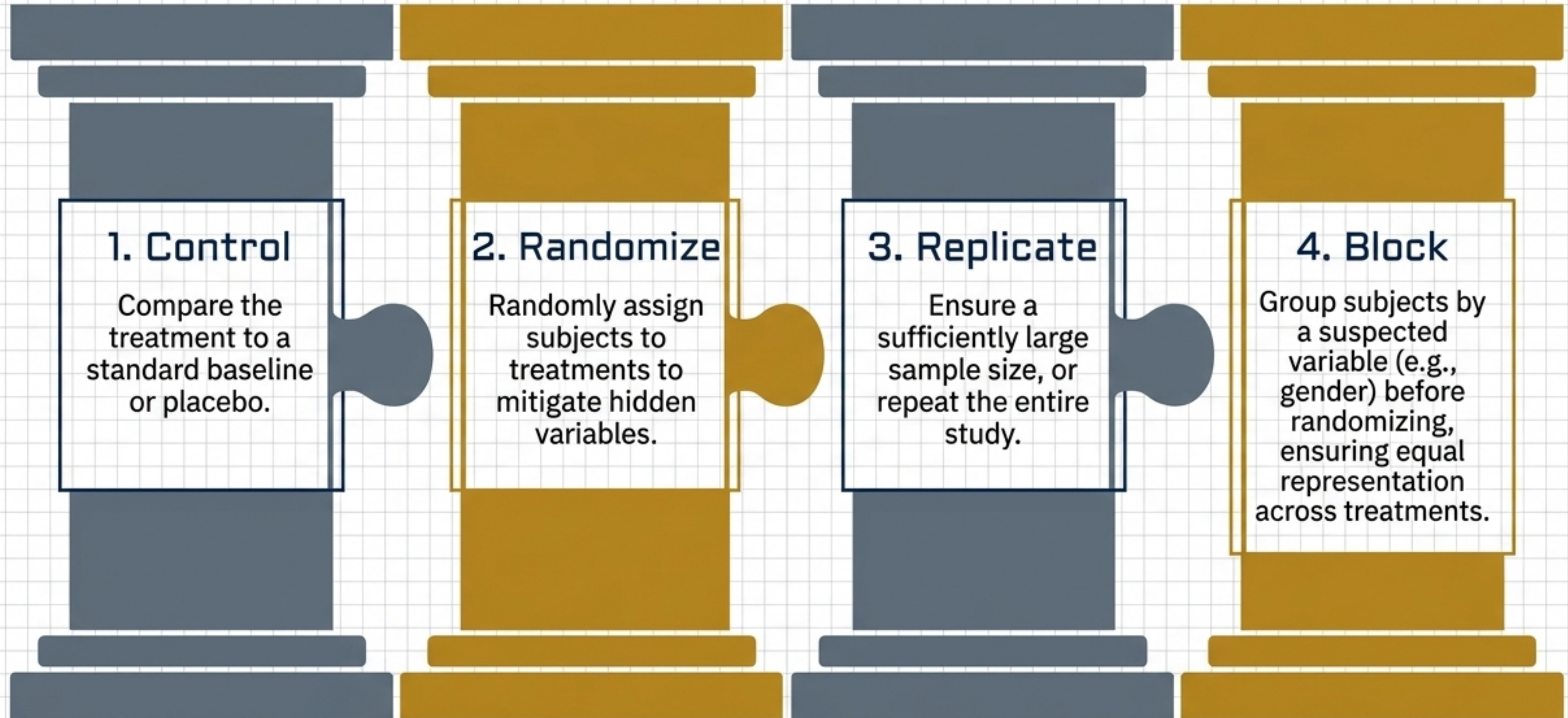
4. Multistage

Like cluster, but we randomly sample within the selected clusters instead of taking everyone.

Real-World Blueprint: Chronic Fatigue Syndrome Trial



Four Principles of Experimental Design



Controlling the Human Element



Placebo: A fake treatment used as a control.



Placebo Effect: Subjects showing improvement merely because they believe they are being treated.



Blinding: Experimental units do not know if they are in the control or treatment group.



Double-Blind: Neither the subjects nor the researchers interacting with them know who is in which group, eliminating observer bias.

Synthesis: The Researcher's Checklist



1. Classify with Precision

Know exactly whether your variables are numerical or categorical.



2. Sample the Whole Soup

Use probability sampling (random, stratified, cluster) to avoid catastrophic bias.



3. Separate Association from Causation

Never assume a trend proves a cause without an experiment.



4. Build an Ironclad Blueprint

Always control, randomize, replicate, and block.

Course Housekeeping



Office Hours:

Wednesdays
11 AM - 12 PM CT

Fridays
3 PM - 4 PM CT

Access:

The weekly class link is pinned at the top of the Blackboard course.

Deadlines:

- Homework 1 is due Sept 3.
- Starting next week, late submissions will incur a penalty. Upload in-class work by midnight tonight.